

2024

JOHN HOLLAND

Optimizing Public Asset Lifecycle Using AWS IoT TwinMaker with John Holland

Learn how tier-one contractor John Holland demonstrated efficient asset development and management using AWS IoT TwinMaker.

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Promotes

data-driven decision making

Visualizes

near real-time analytics

Integrates

di



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Streamlines

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Overview

As a major infrastructure, building, rail, and transport business, [John Holland](#) often faces a complex dilemma: how to manage, integrate, and understand the vast amount of data collected when building an asset. Every beam installed and every room built presents a wealth of information waiting to be deciphered.

To address these challenges, John Holland worked alongside Amazon Web Services (AWS) to develop a proof of concept (PoC) digital twin using [AWS IoT TwinMaker](#)—a service that makes it simpler for developers to create digital twins of real-world systems. The company saw it as an opportunity to unlock value from asset data.

Using AWS IoT TwinMaker, John Holland developed a PoC for the redevelopment of the Sydney Football Stadium and crafted a digital twin of the stadium. With this PoC, John Holland used the capabilities of AWS IoT TwinMaker to demonstrate the value of a digital replica to unlock near real-time analytics and streamline decision-making with a single source of truth.



Opportunity | Using AWS IoT TwinMaker to Visualize and Manage Asset Construction for John Holland

Founded in 1949, John Holland provides leading design, engineering, construction, and asset operations across infrastructure, rail and transport, building, and property in Australia and New Zealand. In the construction industry, being at the forefront of technology is crucial for long-term success. “When building assets, it’s essential to consider what an asset owner requires, not just now, but over the lifespan of the asset,” says Adam Plummer, manager of business technology at John Holland. “For John Holland and our clients, it’s about understanding what they aim to be in the next 5–10 years and using technology as a building block to bring that strategic vision to life.”



In 2020, John Holland spoke about how it wanted to improve the process of building assets and how asset data can offer valuable insights into efficiency, safety, and sustainability. For example, projects that are subject to environmental regulations may be required to monitor and report on factors such as noise, dust, vibration, and air quality. However, large volumes of data can be difficult to capture and decipher. The answer was the digital twin, a digital model of an asset where stakeholders can visualize, track, and understand data from various sources across the lifecycle of a project or asset.

“Creating a digital twin provides the power to reflect the real-world state of an asset, effectively giving executives the ability to understand an asset’s performance at the touch of a button,” says Bastian Uber, chief digital and information officer at John Holland. “That’s what we are striving toward.”

Solution | Working alongside the AWS Team to Innovate on Asset Development Using a Digital Twin

John Holland created a PoC using AWS IoT TwinMaker while redeveloping the Sydney Football Stadium. The arena hosts professional rugby and soccer games and would later serve as one of the venues for the 2023 Fédération Internationale de Football Association (FIFA) Women’s World Cup.

Working with the AWS IoT TwinMaker technical team, John Holland integrated precinct-wide models, Internet of Things sensor data, and third-party systems like noise sensors, temperature sensors, and asset registers, creating a comprehensive digital representation of the stadium. To demonstrate the advantages of using a digital twin, it used AWS IoT TwinMaker to facilitate stakeholder decision-making, monitor room performance, and comply with regulatory noise standards. (See Figure 1: Edge-to-Twin Architecture.)

For the first use case, John Holland created a specific operational twin for stadium management. With this digital replica, asset managers can access vital information, instead of sifting through pages of data. Stakeholders could select a particular section of the stadium in the digital twin and instantly view its associated datasets and operational details. The operational twin has the potential to streamline workflows, helping managers make decisions in near real time rather than relying on manual processes and project reports that are potentially out of date.

The second use case centered on near-real-time room performance condition monitoring. Using AWS IoT TwinMaker, John Holland connected various third-party systems for near-real-time monitoring and managing so that stakeholders could evaluate select rooms in the stadium and make proactive decisions to verify

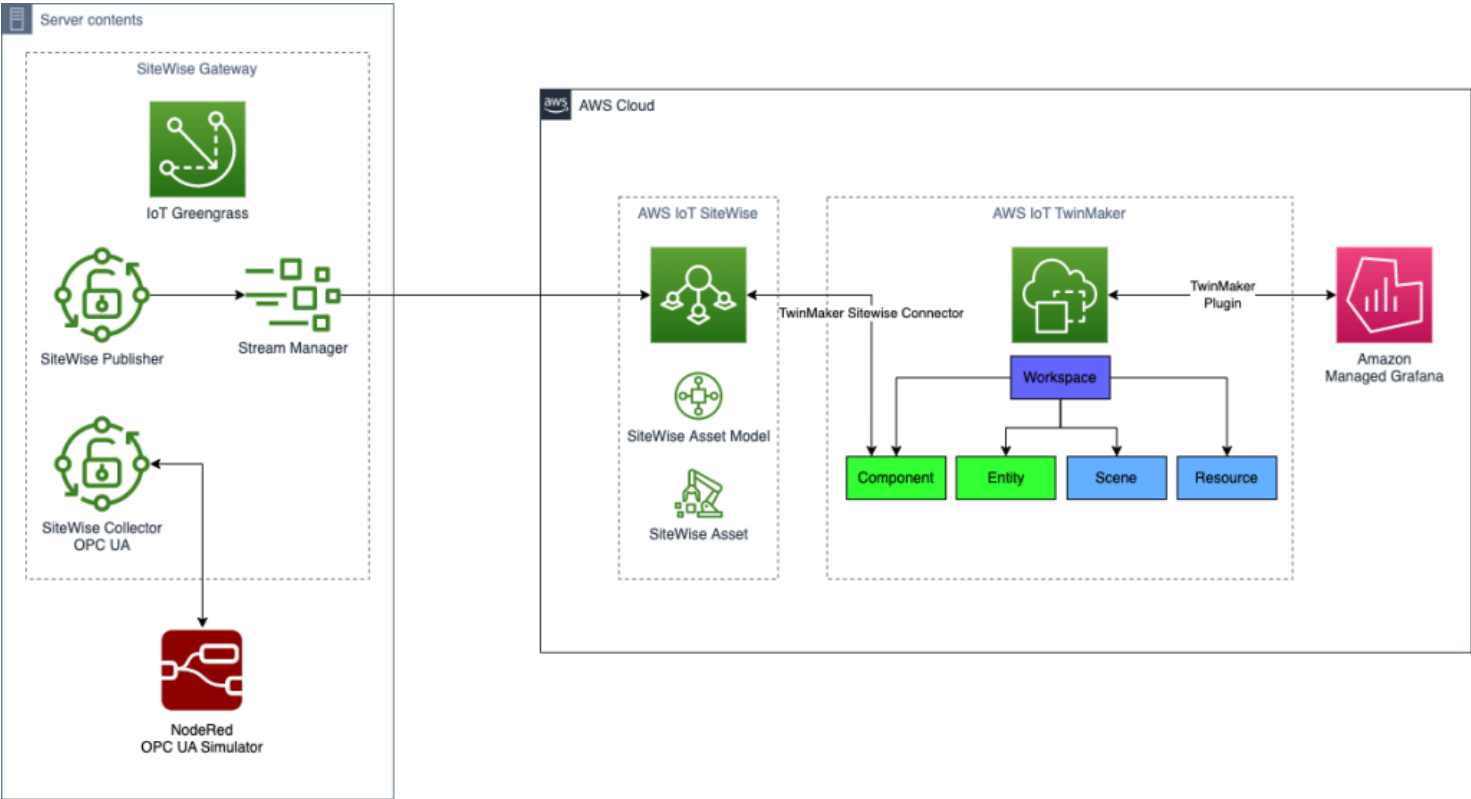


optimal usage and performance. With these capabilities, asset owners could potentially enhance the audience experience and increase the lifespan of each room.

Finally, the third use case focused on stadium noise management. John Holland used a third-party noise-monitoring solution to measure noise levels during stadium construction, helping to manage regulatory noise standards. The digital twin facilitates the visualization of historical noise data, which managers can use to plan events without worrying about potential noise complaints. Equipped with this information, stakeholders can improve community relations and reduce the frequency of noise-related concerns.

Using the digital twin, John Holland demonstrated its ability to bring together many different data points from multiple sources in near real time. Using AWS IoT TwinMaker, the company can present this data in a visual context, helping asset owners and stakeholders make faster, smarter, and better-informed decisions.

Architecture Diagram



Outcome | Driving Change and Decision-Making Using AWS and IoT

John Holland aims to embrace innovative technologies to achieve its goals of being a leading and future-focused asset builder and developer. Using AWS IoT TwinMaker, John Holland demonstrated to stakeholders the ability to connect and visualize near-real-time asset data to deliver new capabilities that are truly future-focused and improve asset construction and operations.

“Using AWS IoT TwinMaker, our PoC helped educate key stakeholders and build confidence around digital twin technology,” says Plunkett. “In the future, I envision a digital twin will assist the optimized development and management of assets and support greater sustainability outcomes across the project lifecycle. This PoC using AWS demonstrated that and aligns with our mission to be an industry leader in technology and digital capabilities.”

About John Holland

John Holland is one of Australia's leading building, infrastructure, rail, and transport companies. It creates innovative and enduring solutions for its customers across multiple industry sectors, delivering complex and city-shaping projects.

AWS Services Used

AWS IoT TwinMaker

AWS IoT TwinMaker makes it easier for developers to create digital twins of real-world systems such as buildings, factories, industrial equipment, and production lines.

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